

Sayan Kumar Roy

AI & MACHINE LEARNING ENTHUSIAST

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👤 SUMMARY

I am a Computer Science graduate with a strong interest in data analysis and machine learning-based solutions. I am committed to solving real-world problems through data-driven decision-making and algorithmic modeling. I have hands-on experience working on several academic and personal projects using tools such as R, SQL, TensorFlow, and Keras. I am eager to work in an environment that offers opportunities for learning, collaborative teamwork, and continuous growth in technical skills.

🎓 EDUCATION

American International University-Bangladesh (AIUB)

2022 — 2026

Bachelor of Science in Computer Science & Engineering (CSE)

CGPA: 3.37 / 4.00

Result Status: Pending / Final Semester

Uttara Model College, Dhaka

2020

Higher Secondary Certificate (HSC) — Science

GPA: 4.75 / 5.00

Rupasdi Brindabon High School, Comilla

2018

Secondary School Certificate (SSC) — Science

GPA: 4.50 / 5.00

</> TECHNICAL SKILLS

Programming Languages

C++, R, Java, C#, HTML, JavaScript, Python

Simulation & Tools

MATLAB, Emu8086, AutoCAD

Research & ML Tools

TensorFlow, PyTorch, NumPy, Pandas, Keras, Google Colab

Professional Competencies

MS Word, Excel, PowerPoint, Power BI, Communication

🌟 CERTIFICATIONS

3 Days Data Analytics Workshop (Tableau & Power BI)

🔗 Dec 2025

BRIT College of Engineering & Technology, UK — via Edulinking Career

Research Paper Writing Masterclass

🔗 Sep 2025

CARSIT (Centre for Applied Research in Software & IT)

RESEARCH EXPERIENCE

Facial Emotion Recognition Framework

ONGOING

Building a Deep Ensemble CNN framework that combines multiple distinct architectural paths to optimize prediction accuracy and robustness for human-computer interaction modules.

Leaf Guard: Plant Disease Detection

ONGOING

Training high-accuracy computer vision classification pipelines on TensorFlow/Keras to scan automated leaf imagery for early diagnosis of crop pathology symptoms.

PORTFOLIO PROJECTS

Amazon Product Reviews Clustering



Developed an unsupervised NLP pipeline to cluster 200,000 reviews using TF-IDF, PCA, and multi-algorithm analytics (K-Means, DBSCAN, Hierarchical) for robust insight reporting.

Handwritten Digit Recognition



Built a CNN system using TensorFlow/Keras recognizing MNIST digits with ~98% accuracy. Integrated live detection feed utilizing OpenCV via webcam framework.

Animal Image Classification (KNN)



Developed an image classifier using K-Nearest Neighbors on animal categories. Implemented OpenCV for structured image preprocessing and 5-fold cross-validation.

Tabular Data Multi-Class Classification



Engineered a custom deep neural network with 4 hidden layers and dropout optimization. Achieved 79% test accuracy on complex categorical multi-class tabular metrics.

Advanced Registration & Login System



Designed a responsive web-based access control interface utilizing HTML/CSS and comprehensive native JavaScript form authentication rules to block data inconsistencies.

Coaching Management System



Programmed a desktop workspace system using C# and PL/SQL. Handles automated report compilation, transaction logs, accounts, and student profiling modules seamlessly.

Online Tutor Management System



Architected a database platform engineered in C# utilizing robust structured query CRUD pipelines to safely synchronize tutor matching, client logs, and calendar schedules.

2D Car Racing Game



Authored an interactive 2D arcade rendering loop in native C++ using basic graphics libraries. Features complex keyboard event pooling, live high score trackers, and polygon boundary collision checks.

REFERENCES

References and supporting transcripts are available upon formal request.